

Phenyo Phemelo Moletsane

Pittsburgh, PA, USA | pmoletsa@andrew.cmu.edu | <https://phenyomoletsane.github.io>

EDUCATION

- Ph.D.**, Carnegie Mellon University, USA **2024–Present**
Human-Computer Interaction, School of Computer Science
• **Advisors:** Amy Ogan, Paulo Carvalho
- Master of IT in Big Data Science**, University of Pretoria, South Africa **2019**
School of Information Technology
- Bachelor of Science in Metallurgy and Materials Engineering** **2013**
Chemical and Metallurgical Engineering
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RESEARCH INTERESTS

Human-Computer Interaction, Learning Sciences, Educational Technologies, Generative AI Learning Tools: Investigating equity-based learning through the lens of multilingualism and local languages inclusion in educational technologies.

RESEARCH PROJECTS

- Yiya Solutions* **2024–Present**
- Yiya Solutions is an offline and remote learning platform that delivers STEM education through radio and basic mobile phones for learners in rural Uganda
- Currently leading the investigation of student engagement and learning with the multilingual Edtech to understand how to better support multilingual learners and design effective multilingual Edtech
 - Conducted a quantitative analysis on a large log data collected from the platform to investigate learning rates, learning outcomes, engagement and demographics
 - Findings from this analysis are published in the CHI' 2026 conference and International Conference of the Learning Sciences 2026, Learning at Scale 2026
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PUBLICATIONS AND PRESENTATIONS

- **Phenyo Phemelo Moletsane**, Michael W Asher, Christine Kwon, Paulo F. Carvalho, and Amy Ogan. 2026. Inclusive Mobile Learning: How Technology-Enabled Language Choice Supports Multilingual Students. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 17 pages. <https://doi.org/10.1145/3772318>. (**Presenter**, CHI'2026).
- **Phenyo Phemelo Moletsane**, Christine Kwon, John Stamper, Amy Ogan, and Paulo F. Carvalho. 2026. Can Multilingual Environments Promote Scalable EdTech? Evidence from a Randomized Controlled Trial. In Proceedings of the Thirteenth ACM Conference on Learning @ Scale (L@S '26), June 29-July 03, 2026, Seoul, Republic of Korea. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3774398.3811600>. (**Presenter**, L@S' 2026).

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- Christine Kwon*, **Phenyo Phemelo Moletsane***, Michael Asher, Dieyu Ouyang, Lingkan Wang, Debbie Eleene Conejo, John Stamper, Paulo Carvalho, and Amy Ogan. 2026. Language Preferences and Practices in Multilingual EdTech: Flexible Primary Language Use with Secondary Language Support. International Conference of the Learning Sciences - ICLS 2026. International Society of the Learning Sciences.
 - Ruiwei Xiao, Qing Xiao, Xinying Hou, **Phenyo Phemelo Moletsane**, Hanqi Jane Li, Hong Shen, and John Stamper. 2026. Do Teachers Dream of GenAI Widening Educational (In)equality? Envisioning the Future of K-12 GenAI Education from Global Teachers' Perspectives. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 20 pages. <https://doi.org/10.1145/3772318.3790908>
 - Christine Kwon, Dieyu Ouyang, Lingkan Wang, Debbie Eleene Conejo, **Phenyo Phemelo Moletsane**, John Stamper, Amy Ogan. 2026. EdTech for Last Mile Learners in the Global South: Navigating Technological and Motivational Learning Insights with Radios and Mobile Phones. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26), April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA <https://doi.org/10.1145/3772318.3790312>

ACADEMIC AWARDS AND SCHOLARSHIPS

MLSS (Machine Learning Summer School) Indonesia Best Proposal Award	2020
Group award for presenting the best proposal about improving education during the COVID19 pandemic.	
International Centre for Theoretical Physics Workshop on Science of Data Science	2019
Secured a travel award to participate in the Science of Data Science workshop.	
ABSA Data Science Bursary Award (University of Pretoria)	2018
For the top 8 performing first year Master's students in Big Data Science (out of 40 students).	
Merit award: Structure and Properties of Engineering Materials and Particulate Systems	2013
Received awards for outstanding performance in the modules.	
Genmin-Adrian Goddard Prize in Extractive Metallurgy (University of the Witwatersrand)	2013
Graduation award - The most deserving and meritorious final year student in Metallurgy/Mineral Process.	
Prof S Luyckx Undergraduate Prize (University of the Witwatersrand)	2013
Graduation award - The 4th year student who has achieved the highest aggregate in Physical Metallurgy courses.	
David Webster Hall Residence Academic Awards (University of the Witwatersrand)	2011–2013
Received top achiever award 3 years in a row, taking 1st position in 2012 and 2013 and 2nd position in 2011 at the university residence out of approximately 217 students.	

WORK EXPERIENCE

Student Researcher	May 2025–Jul 2025
<i>Injini, South Africa</i>	
● Worked with the research team on projects focused on understanding best practices for Blended Learning and Multilingualism in educational technologies	

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- Supported project scoping and developed interview questions
- Conducted interviews with EdTech fellows to understand their solutions, needs, and research questions, and to explore opportunities for collaboration
- Conducted observations at EdTech fellows' partner schools to gather insights and understanding of their EdTech solutions in practice
- Designed and led a workshop to 20 EdTech fellows to introduce evaluation methods for tracking and understanding learning outcomes, including non-experimental, quasi-experimental, and experimental approaches

Clinical Data Scientist

Dec 2020–Jul 2024

FHI Clinical Inc, South Africa

- Created analytical tools for clinical studies as part of risk-based monitoring
- Built centralised statistical monitoring processes and dashboards using R to assist clinical research associates to monitor TB and COVID-19 clinical trials data for errors and outliers.
- Analyzed clinical trial data to highlight areas of potentially high-risk data and detect trends
- Wrote R packages and documentation on Github for the centralised statistical monitoring projects and other projects
- **Verbal communication skills:** Demo presentations on analytical tools to sponsors, stakeholders and non-technical audience
- Activities: Member of the Employment Equity committee

Quantitative Analyst

May 2019–Nov 2020

First National Bank, South Africa

- Analyzed customer data such as demographics, clickstream, spending, credit data to draw insights and build analytical tools
- Developed a data extraction method for a marketing and onboarding campaign which was used to improve the customer engagement and onboarding for new clients
- Built credit risk scoring model for home loans

COMMUNITY WORK

Guest Panelist

May 2026

Black in AI's ELAI Program c

- Spoke on a panel about graduate school applications, AI research pathways, and strategies for succeeding in academia

Guest Speaker

Nov 2025

CMU Research Club Africa

- Delivered a talk on crafting strong and competitive PhD applications for prospective PhD students in Africa, covering academic CV preparation, statement of purpose writing, engaging recommenders effectively, and other key strategies for application success.

Ambassador

Jul 2023–Present

Google's Women Techmakers Program

- Participated in the 'Let's Talk Data' series as a panellist - Educated the audience about data.
- Organised Women Techmakers Breakfast event as part of the DevFest South Africa 2023 - Attended by 20 women

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Machine Learning Engineer

Aug 2019–Dec 2020

Omdena

- Acted as a task leader and collaborated with 30 Omdena volunteers and SafeCity.
 - Created automated safety tips and heat sensitivity algorithms for people reporting sexual harassment on the SafeCity app
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COUSEWORK

- Cognitive Perspectives HCI
- Interactive Data Science
- Design Perspectives HCI
- Language and Thought
- Designing for the Present & Future of Work
- Computer Science Perspectives HCI
- Social Perspectives HCI
- Learning Analytics and Educational Data Science
- Applied Research Methods
- Process & Theory